

Software Requirements Specification

AI-powered Film Photography Contest Platform

Development Team

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Contents

1	User Requirements	2
1.1	Actors	2
1.2	Use Cases	2
1.2.1	Diagram(s)	2
1.2.2	Descriptions	3
2	Functional Requirements	5
2.1	System Functional Overview	5
2.1.1	Screens Flow	5
2.1.2	Screen Descriptions	6
2.1.3	Screen Authorization	6

1 User Requirements

1.1 Actors

Table 1: System Actors and Descriptions

Actor	Description
Participant	Users of the platform who create accounts, register film roll technical specs, upload digitized frames, and submit photos to participate in contests.
Organizer	Entities or individuals hosting contests. Responsible for creating contests, configuring scoring criteria, assigning Judges, and manually reviewing submissions flagged by the AI.
Judge	Photography experts assigned by the Organizer to anonymously evaluate and score submissions based on configured criteria.
Admin	System administrator managing master data, approving Organizer accounts, configuring roles/permissions, and overseeing platform operations.
AI System	An automated background service actor that scans submissions to detect duplicate images, identify AI-generated content (e.g., Midjourney/DALL-E), and assign classification tags.

1.2 Use Cases

1.2.1 Diagram(s)

The PlantUML source code below defines the Use Case Diagram for the platform:

Listing 1: Use Case Diagram Source Code (PlantUML)

```
@startuml
left to right direction
skinparam packageStyle rectangle

actor "Participant" as part
actor "Organizer" as org
actor "Judge" as judge
actor "Admin" as admin
actor "AI System" as ai <<System>>

rectangle "Film Contest Platform" {
    usecase "Account Management (Auth)" as UC1
    usecase "Film Roll & Frame Management" as UC2
    usecase "Photo Submission" as UC3
    usecase "Validity & Fraud Scan" as UC4
    usecase "Contest Creation & Configuration" as UC5
    usecase "Judge Assignment" as UC6
    usecase "Manual Submission Review" as UC7
    usecase "Submission Scoring" as UC8
}
```

```

    usecase "System User Management" as UC9
}

part --> UC1
org --> UC1
judge --> UC1
admin --> UC1

part --> UC2
part --> UC3
UC3 ..> UC4 : <<include>>
ai --> UC4

org --> UC5
org --> UC6
org --> UC7

judge --> UC8

admin --> UC9
@enduml

```

1.2.2 Descriptions

Table 2: Use Case Descriptions Table

No.	Use Case	Use Case Actors	Use Case Description
1	Account Management (Auth)	Participant, Organizer, Judge, Admin	Allows users to register, log in via JWT, reset passwords, and update profile information.
2	Film Roll & Frame Management	Participant	Enables participants to record film specs (stock, camera, ISO) and upload digitized frames with contact sheets.
3	Photo Submission	Participant	Allows participants to select an active contest and submit a frame from an existing film roll.
4	Validity & Fraud Scan	AI System	Automatically triggered upon submission to analyze image similarity, calculate AI generation probability, and return a status verdict.

Table 2: Use Case Descriptions Table

No.	Use Case	Use Case Actors	Use Case Description
5	Contest Creation & Configuration	Organizer	Enables organizers to set up a contest, define timeline milestones, and configure specific scoring criteria (e.g., Grain, Composition).
6	Judge Assignment	Organizer	Enables organizers to assign specific Judge accounts to form the evaluation panel for a contest.
7	Manual Submission Review	Organizer	Allows organizers to inspect AI-flagged entries and make final approval or rejection decisions.
8	Submission Scoring	Judge	Allows judges to anonymously view submissions, slide scores (1–10) per criterion, and provide feedback.
9	System Management	User Admin	Allows administrators to manage accounts, adjust roles, and approve requests to become an Organizer.

2 Functional Requirements

2.1 System Functional Overview

2.1.1 Screens Flow

The PlantUML state diagram below illustrates the navigation flow between system screens across different roles:

Listing 2: Screens Flow Source Code (PlantUML)

```
@startuml
hide empty description
skinparam state {
    BackgroundColor White
}

[*] --> LoginScreen : Open Application

state LoginScreen {
    [*] --> Authentication
}

LoginScreen --> AdminDashboard : Role = Admin
LoginScreen --> OrganizerDashboard : Role = Organizer / Manager
LoginScreen --> JudgeDashboard : Role = Judge / Lecturer
LoginScreen --> ParticipantDashboard : Role = Participant /
    Student

state ParticipantDashboard {
    [*] --> FilmRollsScreen : Manage Photos
    FilmRollsScreen --> FramesDetailScreen : Upload & Select Photo
    ParticipantDashboard --> ActiveContestsScreen : Browse Contests
    ActiveContestsScreen --> SubmitPhotoScreen : Submit Photo
}

state OrganizerDashboard {
    [*] --> ManageContestsScreen : Initialize
    ManageContestsScreen --> SetupCriteriaScreen : Configure
        Scoring
    ManageContestsScreen --> AssignJudgesScreen : Assign
    OrganizerDashboard --> SubmissionsReviewScreen : Review (AI
        Alert)
}

state JudgeDashboard {
    [*] --> AssignedContestsScreen : Select Contest
    AssignedContestsScreen --> JudgingPanelScreen : Evaluate
}

state AdminDashboard {
    [*] --> UserManagementScreen
    [*] --> MasterDataScreen
}
```

```
}
@enduml
```

2.1.2 Screen Descriptions

Table 3: Screen Descriptions Table

No.	Feature	Screen	Description
1	Authentication	Login / Register Screen	User credentials entry, account registration form, and password recovery link.
2	Profile Management	User Profile Screen	Displays personal details, avatar image, security settings, and user activity history.
3	Film Management	Film Rolls & Frames Screen	Grid layout displaying film rolls, enabling technical metadata entry and frame image uploads.
4	Contest Browsing	Contest Browsing Screen	Cards view displaying active contests, countdown timers, and submission guidelines.
5	Submission	Submit Photo Screen	Form to select a frame from an existing film roll, enter title, backstory, and confirm submission.
6	Contest Settings	Contest Settings Screen	Organizer dashboard to initialize contests, set dates, and assign criteria weights.
7	Verification	Submissions Review Screen	Table view of all submissions with red/yellow indicators highlighting AI-flagged entries.
8	Judging	Judging Panel Screen	Dark-mode workspace focused on high-res photos, metadata, and criteria score sliders (1–10).
9	Administration	Admin Dashboard Screen	System stats view with account management tables, role assignments, and system metrics.

2.1.3 Screen Authorization

Note: In the context of this domain, Manager corresponds to the Contest Organizer, Lecturer corresponds to the Judge, and Student corresponds to the Participant.

Table 4: Screen Authorization Matrix

Screen	Admin	Manager	Lecturer	Student
Login / Register Screen	Yes	Yes	Yes	Yes
User Profile Screen	Yes	Yes	Yes	Yes
Admin Dashboard Screen	Yes	No	No	No
User Management Screen	Yes	No	No	No
Contest Settings Screen	View	Yes	No	No
Assign Judges Screen	View	Yes	No	No
Submissions Review Screen	View	Yes	No	No
Judging Panel Screen	No	No	Yes	No
Assigned Contests Screen	No	No	Yes	No
Film Rolls & Frames Screen	No	No	No	Yes
Contest Browsing Screen	View	View	View	Yes
Submit Photo Screen	No	No	No	Yes